Chapter 2a: Fixed Point Concepts

Differential Calculus in Banach Spaces

Based on Hemant Kumar Pathak:
Introduction to Nonlinear Analysis and Fixed Point Theory

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Overview I

Why Differential Calculus in Banach Spaces?

- Classical calculus works in finite-dimensional spaces $\mathbb{R}^n$
- Fixed point theory often involves infinite-dimensional spaces
- Examples:
  - Function spaces: $C[0,1]$, $L^2([0,1])$
  - Sobolev spaces: $W^{1,2}(\Omega)$
  - Sequence spaces: ℓ^p
- **Challenge:** How to extend derivatives to infinite dimensions?
- **Solution:** Two concepts of differentiability
  1. Gâteaux derivative (directional)
  2. Fréchet derivative (full differentiability)

These are essential for fixed point existence theorems!

Banach Space Setup

Definition 1 (Banach Space)

A complete normed linear space $(X, \|\cdot\|)$ where:

- X is a vector space over $\mathbb{R}$ or $\mathbb{C}$
- $\|\cdot\| : X \to \mathbb{R}_{\geq 0}$ is a norm
- All Cauchy sequences converge (completeness)

Example 2

- $\mathbb{R}^n$ with $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$
- $C[0,1]$ with $\|f\|_\infty = \sup_{t\in[0,1]} |f(t)|$
- ℓ^p with $\|x\|_p = \left(\sum_{i=1}^\infty |x_i|^p\right)^{1/p}$

Definition 3 (Dual Space)

The dual space X^* is the space of all continuous linear functionals on X.

Definition: Gâteaux Derivative

Definition 4

Let $F : X \to Y$ be an operator between Banach spaces. F is **Gâteaux differentiable** at $x \in X$ if there exists a continuous linear operator $A : X \to Y$ such that:

$$\lim_{t \to 0} \frac{1}{t} \|F(x + th) - F(x) - tAh\| = 0 \quad \forall h \in X$$

The operator A is called the **Gâteaux derivative** at x, denoted $dF(x)$ or $F'(x)$.

Key interpretation: Limit is taken **along each direction** h separately.

Remark 5

- *The Gâteaux derivative, if it exists, is* **unique**
- *It need* **NOT** *imply continuity of F*
- *It's a directional derivative generalization*

Example: Discontinuous Gâteaux Derivative

Example 6 (Example 3.1)

Let $F : \mathbb{R}^2 \to \mathbb{R}$ be:

$$F(x) = \begin{cases} \frac{x_1^3}{x_2}, & \text{if } x = (x_1, x_2) \neq (0,0) \\ 0, & \text{if } x = (0,0) \end{cases}$$

Then:

- $dF(0) = 0$ (null operator) exists as Gâteaux derivative
- But F is **discontinuous** at $(0,0)$
- For any $h = (h_1, h_2) \in \mathbb{R}^2$:

$$dF(0)h = \lim_{t \to 0} \frac{F(th) - F(0)}{t} = \lim_{t \to 0} \frac{2t^3 h_1^3}{th_2} = 0$$

Gradient and Partial Derivatives

Definition 7 (Gradient)

For a functional $f : X \to \mathbb{R}$ (real-valued), the mapping $x \mapsto f'(x)$ is called the **gradient** of f and denoted ∇f. Thus ∇f is a mapping from X to X^*.

For operators $F : \mathbb{R}^n \to \mathbb{R}^m$ with $F = (f_1, f_2, \dots, f_m)$ and matrix $A = (a_{ij})$:
The Gâteaux derivative $F'(x)$ has matrix representation (Jacobian matrix):

$$F'(x) = \begin{bmatrix} \partial_1 f_1(x) & \cdots & \partial_n f_1(x) \\ \vdots & \ddots & \vdots \\ \partial_1 f_m(x) & \cdots & \partial_n f_m(x) \end{bmatrix}$$

Constant and Linear Operators

Lemma 8

1. *The constant mapping* $F : X \to Y$ *with* $F(x) = a$ *has Gâteaux derivative* $F'(x) = 0$ *(the null operator).*
2. *Any continuous linear operator* $A : X \to Y$ *has Gâteaux derivative* $A'(x) = A$ *for all* $x \in X$*.*

Proof idea for (2):
For linear A:

$$\frac{1}{t}\|A(x+th) - A(x) - tAh\| = \frac{1}{t}\|Ax + tAh - Ax - tAh\| = 0$$

Definition: Fréchet Derivative

Definition 9

An operator $F : X \to Y$ is **Fréchet differentiable** at $x \in X$ if there exists a continuous linear operator $A : X \to Y$ such that:

$$\lim_{\|h\| \to 0} \frac{\|F(x+h) - F(x) - Ah\|}{\|h\|} = 0$$

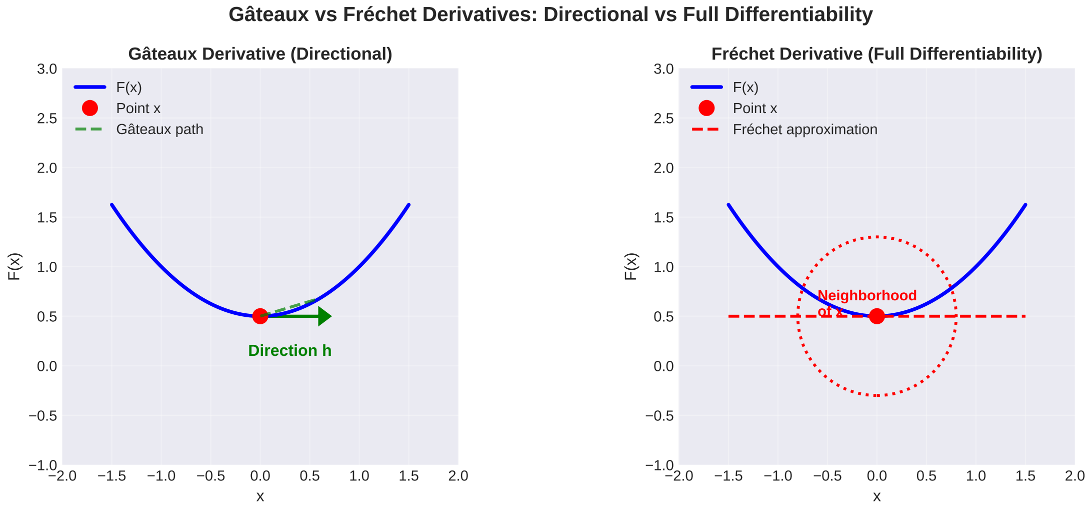
The operator A is the **Fréchet derivative** at x, denoted $dF(x)$ or $F'(x)$.

Key difference from Gâteaux: The limit is taken in a **full neighborhood**, not just along directions.

Remark 10

- *Fréchet differentiability* $\Rightarrow$ *Gâteaux differentiability*
- *Fréchet differentiability* $\Rightarrow$ *Continuity*
- *Converse implications do NOT hold in general*

Comparison: Gâteaux vs Fréchet



Gâteaux: directional limits $\Leftrightarrow$ Fréchet: uniform limit in neighborhood

Example: Fréchet Continuous but Not Gâteaux Continuous

Example 11 (Example 3.2)

Let $F : \mathbb{R}^2 \to \mathbb{R}$ be:

$$F(x_1, x_2) = \begin{cases} \frac{2x_2 \exp(-x_1^2)}{x_2^2 + \exp(-2x_1^2)}, & x_1 \neq 0 \\ 0, & x_1 = 0 \end{cases}$$

Then:

- F is Gâteaux differentiable at $(0, 0)$ with $dF(0) = 0$
- BUT F is **NOT continuous** at the origin
- Reason: along the path $x = (t, t^2)$, $F(t, t^2) \to 2/e \neq 0$ as $t \to 0$

Strong Hemicontinuity

Definition 12

An operator $F : X \to Y$ is **strongly hemicontinuous** at x if for every sequence $x_n \to x$ along a line (i.e., $x_n = x + t_n h$ with $t_n \to 0$), we have $F(x_n) \to Fx$.

Theorem 13 (Theorem 3.5)

If $F : X \to Y$ is Gâteaux differentiable at $x \in X$, then F is strongly hemicontinuous at x.

Proof sketch:

- If $\phi(t) = F(x + th)$ is differentiable at 0 with $\phi'(0) = F'(x)h$
- Then ϕ is continuous at 0
- So $F(x + t_n h) \to F(x)$ when $t_n \to 0$

Mean Value Theorem

Theorem 14 (Theorem 3.1 - Mean Value Theorem)

Let $f : X \to \mathbb{R}$ be a functional with Gâteaux derivative $df(x, h)$ at every point $x \in X$. Then for points $x, x + h \in X$, there exists a constant $\tau \in (0, 1)$ such that:

$$f(x + h) - f(x) = df(x + \tau h, h)$$

Key insight: Extends the classical mean value theorem to Banach spaces!

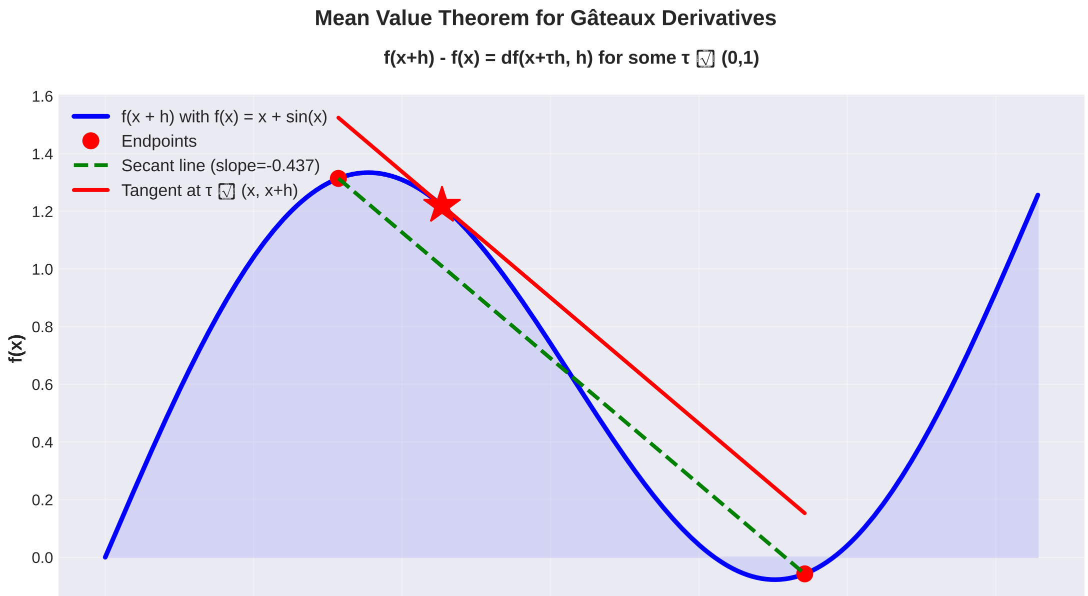
Remark 15

In $\dim X > 1$, *we generally have* **inequality**, *not equality:*

$$f(x + h) - f(x) \neq df(x + \tau h, h)$$

for arbitrary compositions of directions.

Illustration of Mean Value Theorem



Chain Rule for Gâteaux and Fréchet Derivatives

Theorem 16 (Theorem 3.6 - Chain Rule)

Let X, Y, Z be real Banach spaces, $G : X \to Y$ Gâteaux differentiable at x, and $F : Y \to Z$ Fréchet differentiable at $G(x)$. Then $H = F \circ G$ is Gâteaux differentiable at x with:

$$H'(x) = F'(G(x)) \circ G'(x)$$

If $G(x)$ is also Fréchet differentiable, then $H'(x)$ is the Fréchet derivative.

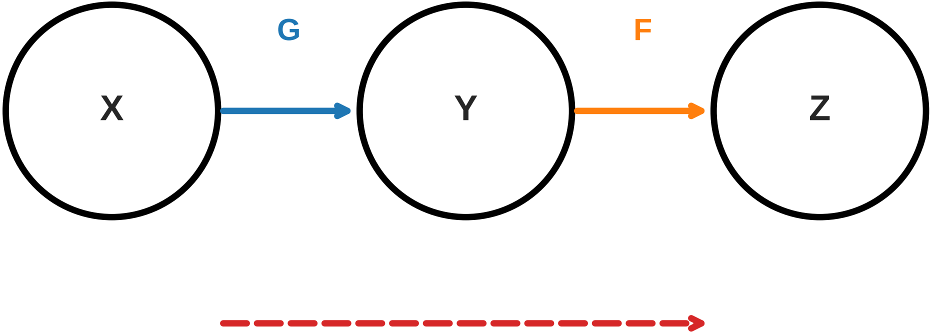
Remark 17

The composition of derivatives:

- $G'(x) : X \to Y$ *(Gâteaux)*
- $F'(G(x)) : Y \to Z$ *(Fréchet)*
- *Product:* $F'(G(x)) \circ G'(x) : X \to Z$

Chain Rule Composition Diagram

Chain Rule for Derivatives: $H = F \circ G$



Proof Strategy for Chain Rule

For $t \neq 0$:

$$\begin{aligned}
&\frac{1}{|t|}\|H(x+th) - Hx - tF'(Gx)G'(x)h\| \\
&\leq \frac{1}{|t|}\|F(G(x+th)) - F(G(x)) - F'(Gx)(G(x+th) - G(x))\| \\
&\quad + |F'(Gx)(G(x+th) - Gx - tG'(x)h)|
\end{aligned}$$

- First term $\to 0$ by Fréchet differentiability of F at $G(x)$
- Second term involves continuity of $F'(Gx)$ and Gâteaux diff. of G

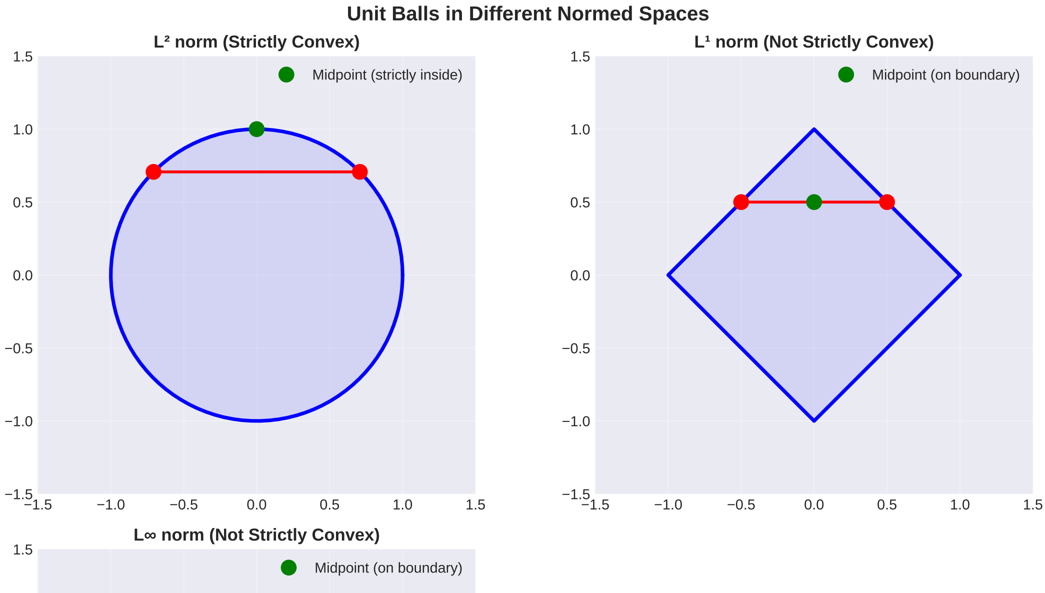
Definition 18 (Strictly Convex Space)

A Banach space X is **strictly convex** if its unit ball $B_X = \{x \in X : \|x\| \leq 1\}$ contains no line segments in its boundary.
Equivalently: if $\|x\| = \|y\| = 1$ with $x \neq y$, then $\left\|\frac{x+y}{2}\right\| < 1$.

Example 19

- L^2 norm on $\mathbb{R}^n$: **strictly convex**
- L^1 and L^∞ norms: **NOT strictly convex**
- $C[0,1]$ space: **NOT strictly convex**

Unit Balls in Different Spaces



Subdifferential Definition

Definition 20

For a functional $f : X \to \mathbb{R}$, the **subdifferential** $\partial f(x)$ at $x \in X$ is the set of all functionals $x^* \in X^*$ such that:

$$f(y) \geq f(x) + \langle x^*, y - x \rangle \quad \forall y \in X$$

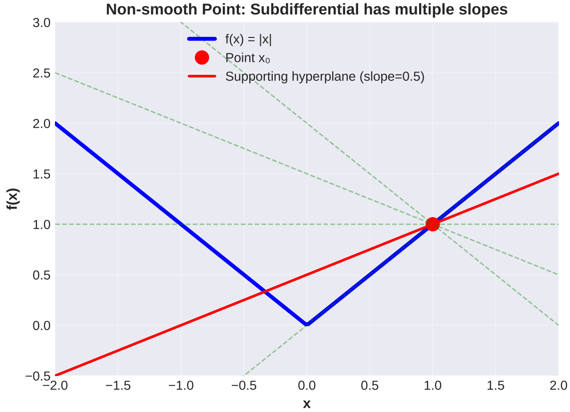
Each $x^* \in \partial f(x)$ is called a **subgradient** at x.

Remark 21

- *For smooth functions:* $\partial f(x) = \{\nabla f(x)\}$ *(singleton)*
- *For non-smooth convex functions:* $\partial f(x)$ *is non-empty, closed, convex*
- *Generalizes the gradient to non-differentiable convex functions*

Subdifferential Illustration

Subdifferential: Supporting Hyperplanes to Convex Functions



Subdifferential Definition

At a point x_0, the subdifferential $\partial f(x_0)$ is the set of all vectors x^* such that:

$$f(x) \geq f(x_0) + \langle x^*, x - x_0 \rangle \quad \forall x \in X$$

Key Properties:

- For smooth functions: $\partial f(x) = \{f'(x)\}$ (singleton)
- For non-smooth functions: $\partial f(x)$ is a set
- For convex functions: always non-empty
- Generalization of gradient to non-smooth functions

Example: $f(x) = |x|$ at $x_0 = 0$

$\partial f(0) = \{z : |z| \leq 1\} = [-1, 1]$

Supporting hyperplanes to convex functions: slopes in subdifferential

Example: Subdifferential of Absolute Value

Example 22 (Norm Functional)

For the norm functional $f(x) = \|x\|$ on a Banach space X:

- At $x \neq 0$: $\partial f(x) = \{x^* \in X^* : \|x^*\|_{X^*} \leq 1, \langle x^*, x\rangle = \|x\|^2\}$
- At $x = 0$: $\partial f(0) = B_{X^*} = \{x^* \in X^* : \|x^*\|_{X^*} \leq 1\}$ (closed ball in dual)

Example 23 (Duality Mapping)

The **duality mapping** $J : X \to X^*$ is defined as:

$$J(x) = \{x^* \in X^* : \langle x^*, x\rangle = \|x\|^2, \|x^*\| = \|x\|\}$$

For strictly convex spaces, J is single-valued (a function).

Conjugate Function and Subdifferentiability

Definition 24

For $f : X \to \mathbb{R}$, the **conjugate function** $f^* : X^* \to \mathbb{R}$ is:

$$f^*(x^*) = \sup_{x \in X} \left(\langle x^*, x \rangle - f(x) \right)$$

Theorem 25 (Theorem 3.21 - Fenchel Duality)

$x^* \in \partial f(x)$ and $x \in \partial f^*(x^*)$ if and only if:

$$\langle x^*, x \rangle = f(x) + f^*(x^*)$$

Key consequence: Subdifferentials are closely linked to duality through conjugate functions.

Subdifferentiability of Convex Functions

Theorem 26 (Theorem 3.22 - Subdifferentiability)

Let f be a lower semicontinuous function on a Banach space X. Then:

$$\mathcal{D}(\partial f) = \mathcal{D}(f)$$

where $\mathcal{D}(\cdot)$ denotes the effective domain (where function is finite).

Remark 27

For convex functions:

- *Subdifferentiable everywhere on interior of domain*
- *May not be differentiable (even Gâteaux)*
- *Subdifferential can be non-empty even at boundary points*

Subdifferential of Norm Functional

Theorem 28 (Theorem 3.23)

The subdifferential of the norm functional $j(x) = \frac{\|x\|^2}{2}$ *on a Banach space* X *is precisely the duality mapping* J*:*

$$\partial j = J$$

Implications:

- The norm is subdifferentiable everywhere
- At non-zero points, subdifferential is unique (for strictly convex spaces)
- At zero, subdifferential is the closed ball in the dual space

Hammerstein Operator

Definition 29

The **Hammerstein operator** $F : L_p[0,1] \to L_p[0,1]$ is defined as:

$$[Fx](s) = \int_0^1 k(s,t) f(t, x(t)) dt$$

where:

- $k(s,t)$ is a kernel function
- $f(t, \cdot)$ is nonlinear in the second argument

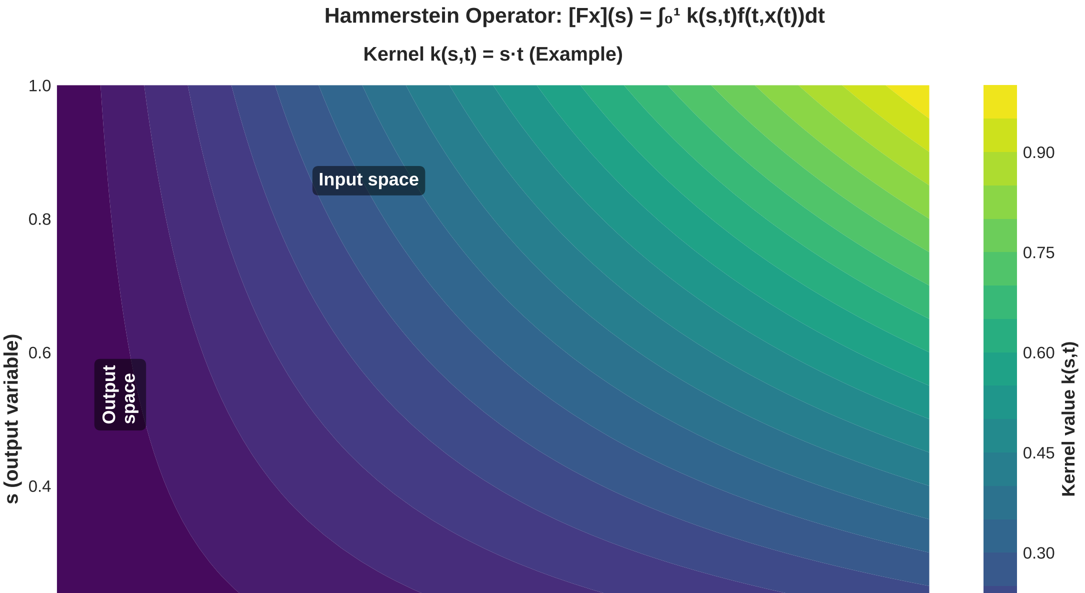
Example 30 (Gâteaux Differentiability)

Under suitable conditions on k and f:

$$dF(x)h = \int_0^1 k(s,t) \frac{\partial f}{\partial u}(t, x(t)) h(t) dt$$

The Gâteaux derivative is the integral operator with kernel and partial derivative.

Hammerstein Operator Visualization



Implicit Function Theorem Application

Problem 31

Suppose $F : X \times Y \to Z$ is continuously Fréchet differentiable with $F(x_0, y_0) = 0$. If $D_y F(x_0, y_0)$ is invertible, then:

- *There exist neighborhoods $U \subset X$, $V \subset Y$*
- *And a unique continuous function $\varphi : U \to V$ with $\varphi(x_0) = y_0$*
- *Such that $F(x, \varphi(x)) = 0$ for all $x \in U$*

Fixed point perspective:

- Can be used to prove existence of solutions to equations
- Foundation for continuation methods in fixed point theory
- Links differential calculus to implicit function existence

Example: Hilbert Space Norm Functional

Example 32 (Example 3.21)

Let X be a real Hilbert space and $f(x) = \|x\|$ for $x \in X$.

- For $x \neq 0$:

  $$\nabla f(x) = \frac{x}{\|x\|}$$

  Gâteaux differentiable with this gradient.

- At $x = 0$:
  - NOT differentiable (singular point)
  - **Subdifferentiable**: $\partial f(0) = \{x^* \in X^* : \|x^*\| \leq 1\}$
  - This is the closed unit ball in the dual

Indicator Function and Normal Cones

Example 33 (Example 3.23 - Indicator Function)

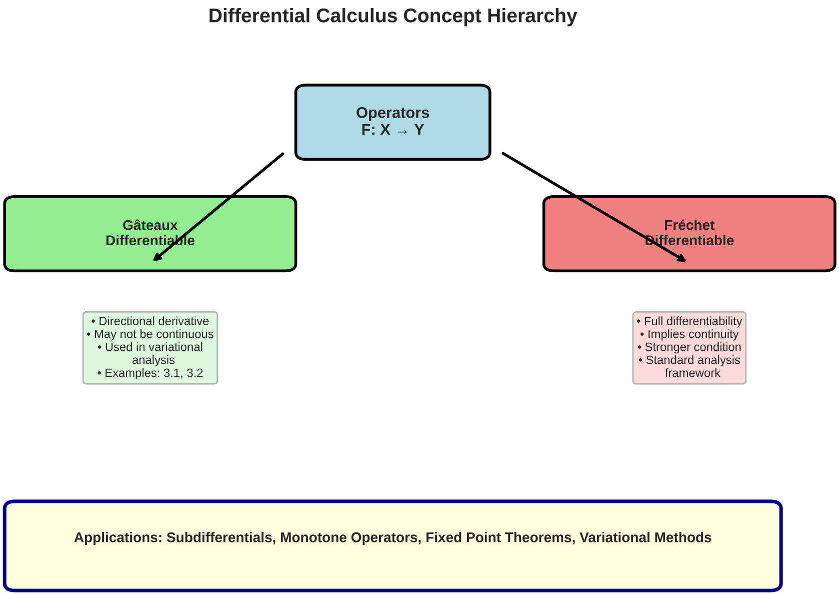
For a convex subset C of a Hilbert space X:

$$I_C(x) = \begin{cases} 0, & \text{if } x \in C \\ \infty, & \text{if } x \notin C \end{cases}$$

At boundary point $x_0 \in \partial C$:

$$\partial I_C(x_0) = \{x^* \in X : \langle x^*, x - x_0 \rangle \leq 0 \text{ for all } x \in C\}$$

This is the **normal cone** to C at x_0.


Differential Calculus Concept Hierarchy
Operators
F: X → Y
Gâteaux
Differentiable
Fréchet
Differentiable
• Directional derivative
• May not be continuous
• Used in variational analysis
• Examples: 3.1, 3.2
• Full differentiability
• Implies continuity
• Stronger condition
• Standard analysis framework
Applications: Subdifferentials, Monotone Operators, Fixed Point Theorems, Variational Methods

Key Definitions Summary

Concept	Banach Space	Related To
Gâteaux Derivative	Directional	Variational Methods
Fréchet Derivative	Full Differentiability	Standard Analysis
Strong Hemicontinuity	Intermediate	Fixed Point Theorems
Subdifferential	Non-smooth Functions	Monotone Operators
Duality Mapping	Dual Space Structure	Banach Space Geometry

Connection to Fixed Point Theory

Why this matters for fixed points:

1. **Gâteaux derivatives:** Used in variational formulations
   - Fixed point equations as variational problems
   - Energy functionals and minimization
2. **Subdifferentials:** Handle non-smooth fixed point maps
   - Proximal operators in iterative methods
   - Monotone inclusion problems
3. **Monotone operators:** Defined via subdifferentials
   - F is monotone if $\langle F(x) - F(y), x - y \rangle \geq 0$
   - Essential for fixed point theorems

Numerical Example: Computing Derivatives

Problem 34

Consider the operator $F : \mathbb{R}^2 \to \mathbb{R}$ *with* $F(x,y) = x^2 + xy + y^2$*.*
Compute the Gâteaux derivative at $(1,1)$ *in direction* $h = (1,0)$*.*

Solution:

$$\begin{aligned} dF(1,1)(1,0) &= \lim_{t\to 0} \frac{F(1+t,1) - F(1,1)}{t} \\ &= \lim_{t\to 0} \frac{(1+t)^2 + (1+t) + 1 - 3}{t} \\ &= \lim_{t\to 0} \frac{1 + 2t + t^2 + 1 + t + 1 - 3}{t} \\ &= \lim_{t\to 0} \frac{3t + t^2}{t} = 3 \end{aligned}$$

The Jacobian is $\nabla F = (2x + y, x + 2y)$, so $\nabla F(1,1) = (3,3)$.

References and Further Reading I

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- Rockafellar, R. T., & Wets, R. J. B. (2009). *Variational Analysis*. Springer.

Key Topics to Explore Next:

- Monotone Operators (Chapter 4 in Pathak)
- Fixed Point Theorems (Chapter 5 in Pathak)
- Topological Degree Theory (Chapter 6 in Pathak)